Statement of Purpose

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I am applying to pursue research in systems at the The University of Utah. During my master's studies at the University of Illinois Urbana-Champaign, I developed a strong interest in this field through my involvement in systems and software research groups and by taking advanced Computer Science courses. For my Ph.D., I aim to focus on designing reliable and efficient software systems. My current research interests include operating systems and machine learning for systems.

Given my growing interest in operating systems research, I was fortunate enough to work at the UIUC xLab. I primarily worked in two projects of Prof. Tianyin Xu's research group.

Making Linux fly. The reliability of code coverage tools is critical for software certified by the Federal Aviation Administration, which relies on metrics like line, branch, and MC/DC coverage to ensure software readiness. In collaboration with Boeing, this project first aims to evaluate the reliability of the coverage tools `gcov` and `llvm-cov` for Debian packages such as apache2, grep, and net-tools. Through our research, I identified several bugs in these tools. For instance, line coverage miscounts occur in `for` loops with empty predicates or when using true from `<stdbool.h>` as the predicate. Additionally, storing the result of a ternary operation in a static global variable causes errors. These bugs were reported to and confirmed by the developers.

We also observed inconsistencies in coverage values across repeated runs, stemming from non-deterministic factors. To investigate, we disabled address space layout randomization in the kernel. This adjustment led to consistent line coverage results for the "stackvma.c" file in the grep package, which involves virtual memory addressing. To improve the reliability of these tools, we developed oracle versions that account for more cases of non-determinism where reasoning is possible. We anticipate that our findings will contribute to the refinement of these tools, enhancing their reliability. We plan to publish our work at a top software engineering conference early next year.

Rust for systems. I am exploring Rust for systems, specifically designing with safe Rust to eliminate the need for eBPF's in-kernel verifier. I contributed to redesigning a machine problem focused on safe eBPF, originally developed by our group, which was incorporated into the undergraduate operating systems course at the Siebel School of Computing and Data Science. Weekly paper discussions were always fun, we analyzed real world systems like *Discord* and how switching to Rust solved their latency spike issues every two minutes due to garbage collectors used in other languages. We also studied some research systems like *Theseus*, *RedLeaf* and how they rely on using Rust compile time checks.

For my advanced IoT class, I am particularly proud of is designing a meta scheduler from scratch, specifically to address limitations in current IoT-specific operating systems. While many IoT-targeted OS solutions exist, I felt that it is imperative to modify the current de-facto Linux Kernel. This led me to develop a priority-based meta-scheduler designed for small embedded systems with a fixed set of tasks. My scheduler maintains a process list

and dynamically controls the "niceness" of processes based on user-defined priority levels. It interfaces with the Linux Completely Fair Scheduler (CFS), adjusting task priorities to ensure high-priority tasks receive precedence. When a high-priority process arrives, the scheduler increases the "niceness" of other processes, then returns them to their initial value once the high-priority process completes. Additionally, the recent news of real-time support being added to the Linux Kernel reinforces the relevance of this work and motivates me to further explore scheduling for real-time systems.

Despite the harsh Illinois winters, I enjoyed going outside to set up Raspberry Pis and experiment with wireless communication. My project partner and I built a wireless mesh network using six Raspberry Pi nodes running the BATMAN protocol, testing it in an outdoor environment. We focused on measuring the time it takes for the network to recover after introducing packet loss with different routing metrics. One intriguing finding was that bandwidth decreased about one minute after packet loss began. After researching this phenomenon, we hypothesized that the delay was due to the sliding window mechanism in BATMAN's routing algorithm, which buffers packet loss information before it impacts the network.

Before my master's program, I worked as a software engineer in the research division at Times Internet, where I focused on deploying machine learning models in production environments. We experimented with open-source embedding models, fine-tuning them for our use case, which led to the development of a Contextual Advertisement Engine based on the BERT-MiniLM model. This engine improved our Click-Through Rate metric by 20%. However, what excited me most was the technical challenge of parallelizing 100 million daily inference calls across multiple GPUs. This industry experience exposed me to the coordination required for large-scale systems and the tradeoffs involved in balancing performance and resource allocation.

At Utah, I am eager to work with Professors Anton Burtsev and Ryan Stutsman. Professor Burtsev's research on isolation, virtualization, and heterogeneous hardware aligns with my interests, and I look forward to contributing to the **Mars Research Group**. Professor Stutsman's innovative OS projects, including *Splinter* and *Cinde*, are equally inspiring. I am also excited about potential collaborations with the **Flux Research Group**, advancing large-scale distributed systems, and with Professor **Rohan Basu Roy**, whose work *The Hidden Carbon Footprint of Serverless Computing* highlights the importance of energy-aware systems.

The practical aspects of building systems with **minimal black box** complexity and being able to interpret their behavior is what makes systems research so fascinating. I believe that at Utah, I will have the opportunity to collaborate with energetic peers and faculty who are passionate about conducting outstanding research. My long-term goal is to explore new research directions and eventually apply for faculty positions in teaching and research after completing my graduate studies.